

Conditional GAN

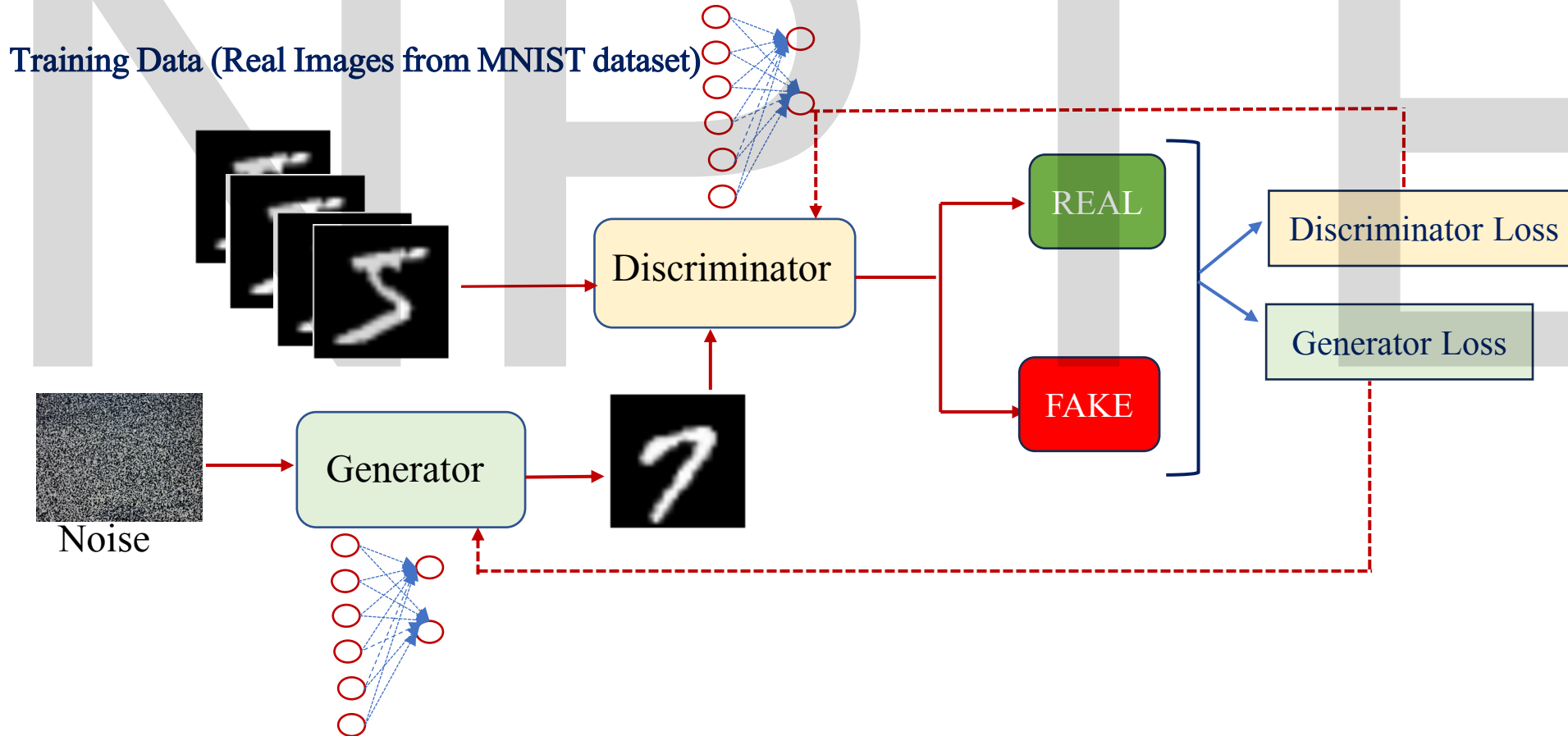
Contents

- ✓ Quick review on GAN
- ✓ Motivation for Conditional GAN
- ✓ Introduction to Conditional GAN (cGAN) architecture
- ✓ Training in cGAN
- ✓ Objective function in cGAN
- ✓ Key difference between Vanilla GAN and cGAN
- ✓ Applications

Quick Review on Working of GAN

- Generative Adversarial Network (GAN) is a deep learning model consisting of a Generator and a Discriminator trained in an **adversarial** manner.
- Generator creates fake samples while the discriminator distinguishes between real and fake samples.
- GANs are trained through a minimax game where the generator and discriminator compete to improve each other.

involving two entities that are in opposition or competition with each other

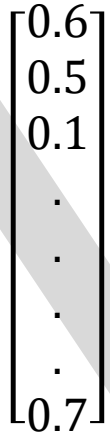


What is Random Noise?

Latent Space:

Latent space refers to an abstract, often lower-dimensional representation of data that captures its essential features and underlying patterns

Random noise is a vector in the latent space. It's a latent vector Z


$$\begin{bmatrix} 0.6 \\ 0.5 \\ 0.1 \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ 0.7 \end{bmatrix}$$

$Z \sim N(0,1)$ — Standard Normal (Gaussian) noise

Each component of Z is drawn independently from a normal distribution with:

Mean = 0

Variance = 1

If it is 100 dimensions, then it has 100 random numbers



But no control on what gets generated

Latent space does not have explicit semantic meaning in standard GANs

Suppose a GAN is trained with
MNIST dataset (0-9)

Now you sample:

$z_1 \rightarrow G(z_1) \rightarrow 3$

$z_2 \rightarrow G(z_2) \rightarrow 7$

$z_3 \rightarrow G(z_3) \rightarrow 1$

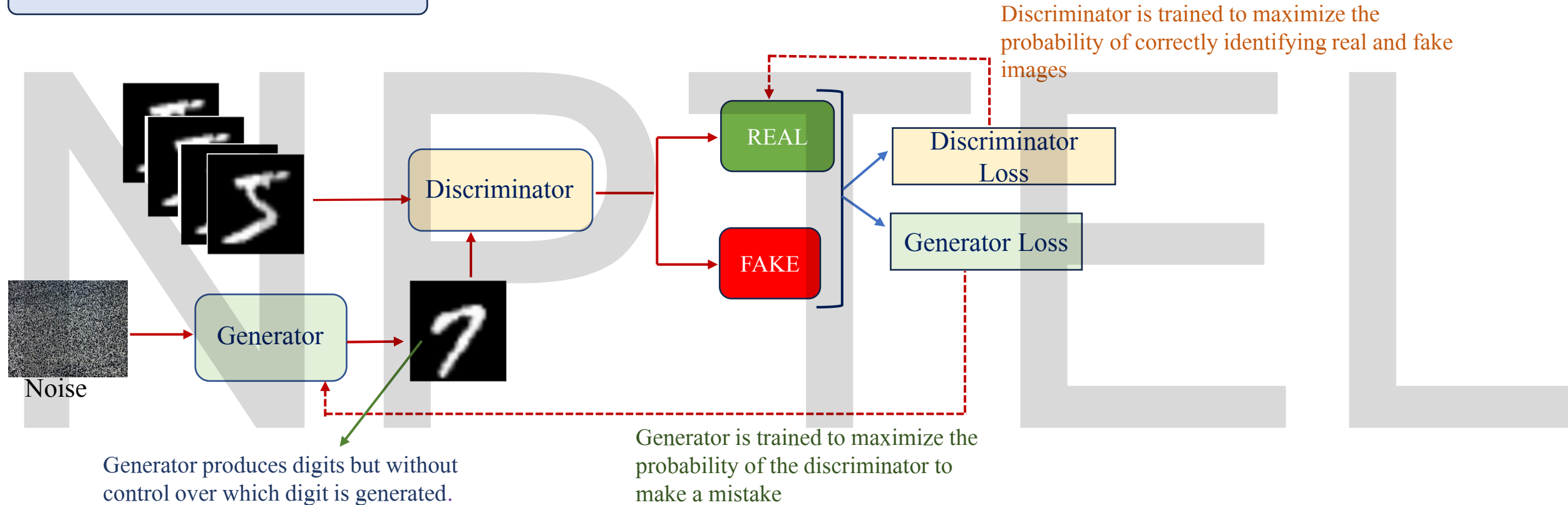
Can I generate a digit 5 on demand?

Answer: No

We don't know which z gives digit 5

Motivation for Conditional GAN (cGAN)

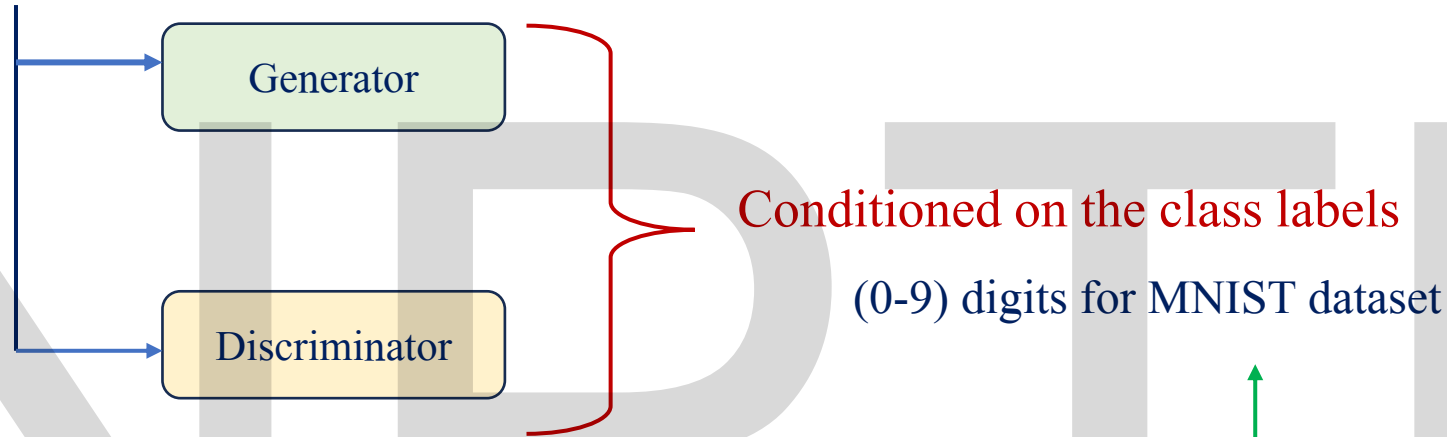
Limitation of Standard GANs



Conditional GANs enable controlled and guided data generation

What is Conditional GAN

Conditioning your GAN using additional information



When we are training our Generator and Discriminator, we will provide **class labels** along with the images.

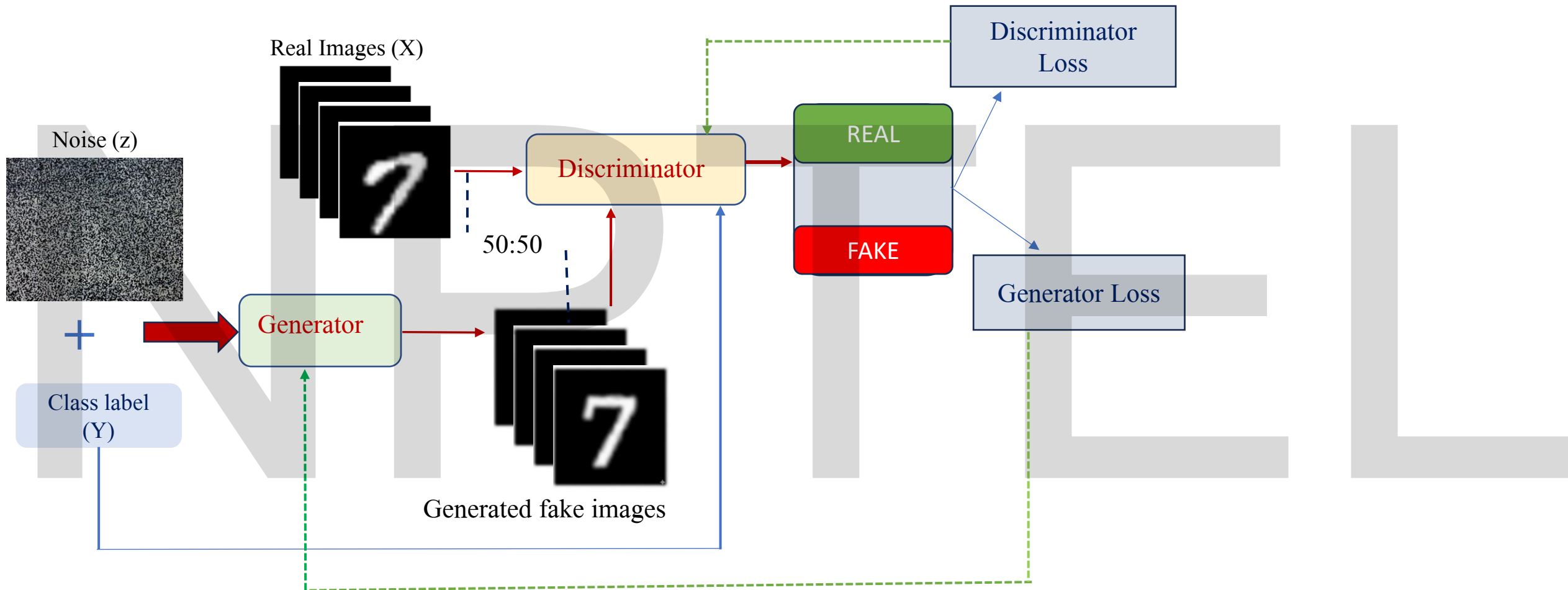
GANs can also be conditioned on

- ✓ Other modalities (image to image translation)
- ✓ Text prompts.

Ref:

*Mehdi Mirza, Simon Osindero , **Conditional Generative Adversarial Nets**, arXiv:1411.1784 [cs.LG]*

Working of a Conditional GAN



Training Loop in Conditional GAN

Step 1:

Pick a real image
(x)



Pick its condition
(y)

Label=7

Discriminator
 $D(x, y)$

Discriminator
sees:
 $D(x_{fake}, y)$

Step 2:

Sample random noise
(z)

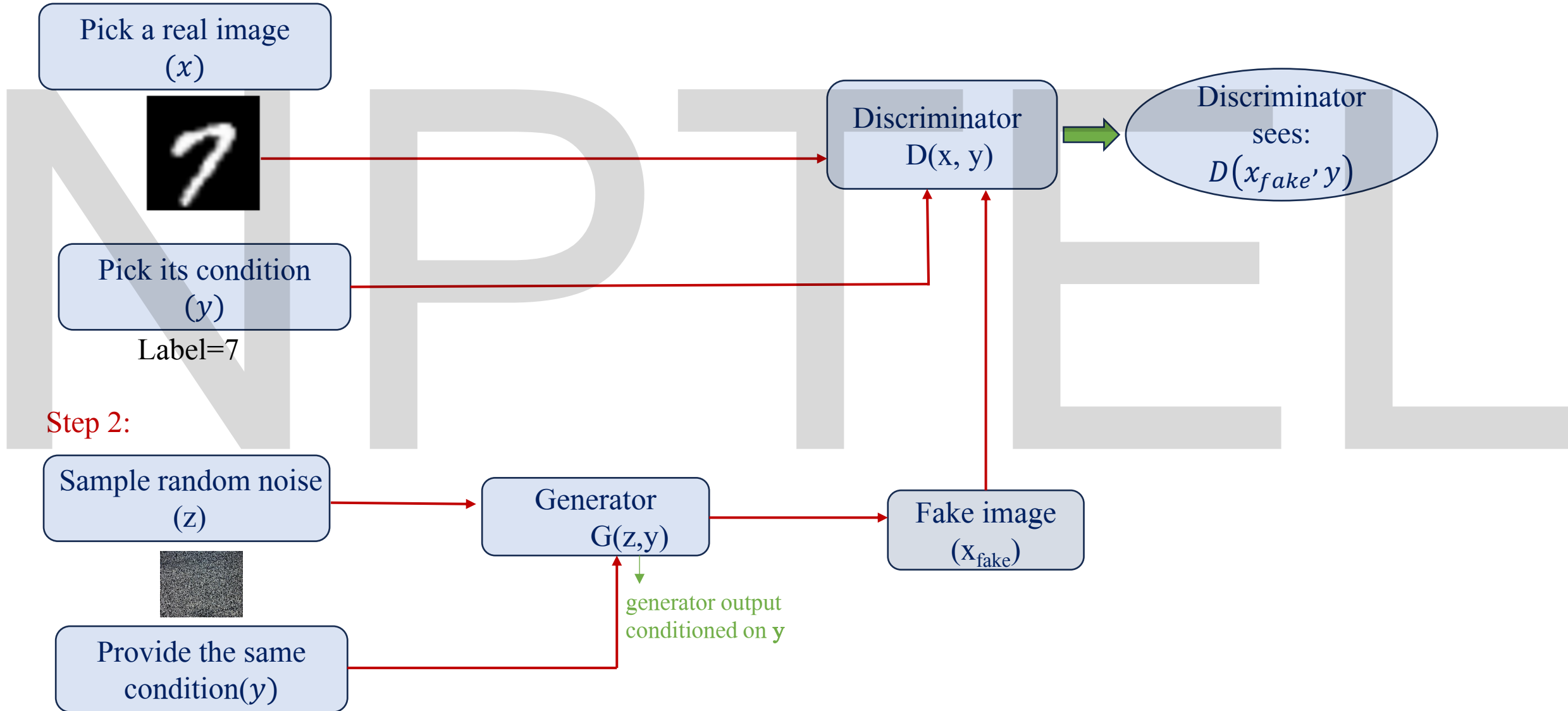


Provide the same
condition(y)

Generator
 $G(z, y)$

Fake image
(x_{fake})

generator output
conditioned on y



Training Loop in Conditional GAN

Step 3:

Train Discriminator

$$D(x, y) \rightarrow 1(\text{real})$$

$$D(G(z, y), y) \rightarrow 0(\text{fake})$$

The discriminator learns two things:

- Does the image match the condition?
- Is it real?

Step 4:

Train Generator

$$D(G(z, y), y) \rightarrow 1$$

The generator learns two things:

- Given label y , generate an image that looks real AND
- Matches y

Step 5:

Repeat

- ✓ Generator improves image quality
- ✓ Discriminator improves detection
- ✓ Eventually generator learns to produce realistic, label-consistent samples.

Discriminator Loss in cGAN

Goal of Discriminator:

- ✓ Predict real pairs as real
- ✓ Predict fake pairs as fake

Real Loss

measures how well D correctly
classifies real data as real

$$\mathcal{L}_D^{\text{real}} = -\mathbb{E}_{x \sim p_{\text{data}}(x)} [\log D(x, y)]$$

expectation (average)
over the real dataset

true data distribution

$D(x, y) = P(\text{real} | x, y)$

Output is a probability scalar
(0–1)

Discriminator wants: $D(x, y) \rightarrow 1$

Negative sign is given for minimizing the loss

Discriminator Loss in cGAN

Fake Loss

Measures how well *discriminator* can detect generated data as fake

$$\mathcal{L}_D^{\text{fake}} = -\mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z, y), y))]$$

Expectation (average)
over noise samples

Discriminator wants: $D(G(z, y), y) \rightarrow 0$

Total Discriminator Loss

$$\mathcal{L}_D = -\mathbb{E}_{x \sim p_{data(x)}} [\log D(x, y)] - \mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z, y), y))]$$

$G(z, y)$: Produces a fake sample conditioned on y

$D(G(z, y), y) = P(\text{real} \mid \text{generated sample}, y)$

Generator Loss in cGAN

Goal of Generator

Generator tries to fool the discriminator by producing samples that look real for that specific condition.

Standard Generator Loss

$$\mathcal{L}_G = \mathbb{E}_{z,y} [\log(1 - D(G(z,y), y))]$$

But vanishing gradient problem occurs if this generator loss is taken

Practical Non
Saturating
Generator Loss

$$\mathcal{L}_G = -\mathbb{E}_{z \sim p(z)} [\log D(G(z, y), y)]$$

Generator wants to maximize this value

$$D(G(z, y), y) \rightarrow 1$$

probability score that the Discriminator assigns to the Generator's fake creation.

Expectation (average)
over noise samples

Negative term is added to minimize the loss

Measures how well the generator can fool the discriminator

Objective function in cGAN

In cGAN, the overall objective is a **min–max game** between Generator G and Discriminator D , both conditioned on y :

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{data(x)}} [\log D(x, y)] + \mathbb{E}_{z \sim p_z(z)} \log(1 - D(G(z, y)))$$

Discriminator tries to maximize its ability to distinguish real from fake while Generator tries to minimize the Discriminator's success.

Overall objective function that both networks are competing to optimize.

encourages the discriminator to classify real samples correctly

pushes the generator to produce samples that the discriminator classifies as real.

- ✓ As training progresses both the generator and discriminator improve.
- ✓ This adversarial process results in the generator producing more realistic data conditioned on the input information.

Key Difference between Vanilla GAN and Conditional GAN

Vanilla GAN



Generator's output depends only on the random noise i.e. $G(z)$



$D(x)$: Probability that input x is **real** (from data)



No control on generation

Conditional GAN

Generator's output depends on both random noise and the class label i.e. $G(z, y)$

$D(x, y)$: Probability that input x is **real** (from data) and it matches with condition y

Controlled generation

Applications of Conditional GAN

Image to Image Translation



Sketch to original
image



Day time to night
time



Text to Image Translation

Create a red bird with blue wings



Generate high-resolution images
from low-resolution inputs



Convert photo → painting style (Van Gogh style.)



NPTEL

Summary

Next Session:

*Image to image Translation
Pix2Pix GAN*